



Investigation report

Haul truck and light vehicle collision, Bindarra Gold Mine - Executive brief

Drafted 04/10/2026

INVESTIGATION WORKFLOW SUITE

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What happened

On the night shift of 11 to 12 August 2026, a 220 t class rigid haul truck (HT-407) was working in the Stage 4 open pit at Bindarra Gold Mine. The pit returned to work after its meal break at 02:22 local time, and the truck was on its second cycle after the break, travelling empty down the main ramp. A grade control technician employed by a survey subcontractor, on his third night shift at the mine, was alone in a light vehicle (LV-118). He was taking levels at a dewatering bore at the end of the RL 312 dewatering spur, a 240 m access road that joins the main ramp. The spur had been opened to light vehicles on 2 August 2026, and the construction sign at its mouth was removed that day with nothing put in its place. Nobody else on the shift knew the technician was in the pit. At about 02:45 he finished his work and drove out along the spur, reaching the ramp at about 02:47.

At 02:47:06 the truck was travelling at 37 km/h, 41 m above the intersection, as the light vehicle came into view beyond a crest in the ramp. About 1.4 seconds later the truck operator applied the brakes and the retarder (an additional braking system on the truck) at full strength and steered right. The truck's collision awareness system detected the light vehicle at 02:47:09 at 21.4 m. The intersection lay inside a mapped zone (geofence) in which alerts were suppressed, so nothing was shown in the cab. Neither driver made a radio call on the pit channel before entering the intersection. The light vehicle entered the ramp without stopping. At 02:47:12 the front left of the truck struck its right side at 24 km/h, 6.2 m into the ramp. The light vehicle was pushed 18.4 m down the ramp and rolled onto its left side against the inner windrow (a bank of material along the road edge). The truck stopped at 02:47:14. Dispatch stopped all Stage 4 traffic at 02:48:01 and the emergency response team was paged at 02:48. The first response vehicle reached the scene at 02:54, and a second responder arrived from the village with a hydraulic rescue tool at 03:07. The driver, who remained conscious throughout, was freed through the rear of the vehicle at 03:14. He had a fractured left collarbone, three fractured left ribs and a small pneumothorax (air in the chest around the lung). He was flown to the metropolitan trauma centre at 07:40. The light vehicle was destroyed, and its seatbelt and roll-over protection structure both showed signs of loading. The truck was damaged at the front left and returned to service on 14 August. The Stage 4 pit stood down for 9 hours 15 minutes and the main ramp below RL 340 was closed for 22 hours.

Why it happened

The collision happened because both vehicles arrived at the intersection with neither driver aware of the other. The investigation found this extremely likely, and found it extremely likely that it led directly to the truck striking the light vehicle. Four conditions at the intersection combined to leave each driver unaware of the other.

The first was the layout of the intersection itself. The spur met the main ramp beyond a crest, giving 41 m of sight line against the minimum of 120 m set in the mine plan. It is very likely that this short sight line contributed to neither driver knowing the other was there until too late. The second was that the light vehicle entered the ramp from the spur without stopping and without establishing where the truck was. It is very likely that this contributed. The third was that neither driver made a positive call (a radio call announcing the vehicle's position and intention) on the pit channel before entering the intersection. It is likely that this contributed. The fourth was that the truck's collision awareness alerts were suppressed inside a geofence that had been extended to cover the new intersection. As a result, the system detected the light vehicle but showed nothing in the cab. It is likely that this contributed.

Several of these conditions traced back to how the spur was brought into use. The spur was cut and opened to traffic under a works order, and no management of change (a formal process for assessing a change before it is made) was raised. It is very likely that this led to the substandard sight line at the intersection. It is also very likely that it left the new intersection outside the site's radio call requirements. Those requirements applied only at intersections named on a list, and the list was rewritten once a year. It is very likely that this arrangement contributed to neither driver making a positive call. Both of these are safety issues.

Behind the change process sat the traffic management guidance. That guidance left the trigger for reviewing traffic arrangements undefined and left the identification of interaction zones to each site. It is likely that this led both to the spur being opened without a management of change and to the radio call arrangement. This is a safety issue.

A separate safety issue concerned the light vehicle driver's preparation for pit driving. The contractor's induction was accepted as equivalent to the mine's own. The site supplement and the pit driving assessment were never verified. It is likely that this contributed to the light vehicle entering the ramp without stopping and without establishing the truck's position.

The investigation also examined whether dust on the ramp reduced the truck operator's ability to see the light vehicle. It could not settle this, and assessed it as about as likely as not. Dust was present in the right place at the right time. However, the operator saw the light vehicle at the distance the ramp geometry predicts for clear air, which leaves dust with nothing to explain. Its effect on detection was not established either way, and the available information cannot establish it.

What we found

A finding stated without a likelihood term was established as fact.

Contributing safety factors

- Collision awareness alerts were suppressed inside a geofence that had been extended to cover the new intersection, and it is likely that this contributed to the collision. (Safety issue)
- The traffic management guidance left the review trigger undefined and left the interaction zones to the site, and it is likely that this contributed to the occurrence through the way the mine opened new roads and nominated its radio call intersections, although the mine's plan met the guidance on its face. (Safety issue)
- Because the access spur was cut and opened to traffic under a works order, no management of change was raised, and this very likely contributed to the occurrence. (Safety issue)
- The spur met the ramp beyond a crest, giving 41 m of sight line against a plan minimum of 120 m, and it is very likely that this contributed to the collision.
- Positive radio calls were required only at listed intersections, and the list was rewritten once a year, so the new intersection could not be added between reviews, and this very likely contributed to the occurrence. (Safety issue)
- Neither operator made a positive call on the pit channel before entering the intersection, which matched what the plan required and what was practised across the pit, and it is likely that this contributed to the collision.
- It is very likely that the contractor induction was accepted as equivalent, with the site supplement and pit driving assessment never verified, and it is likely that this contributed to the occurrence. (Safety issue)
- With no sign, give-way control or pit driving assessment to guide him, the light vehicle driver entered the ramp from the spur without stopping and without establishing the truck's position, and it is very likely that this contributed to the collision.
- It is very likely that both vehicles arrived at the intersection with neither operator aware of the other, and very likely that this led to the collision.
- The haul truck's front left struck the right side of the light vehicle at 24 km/h, and this impact was how the two vehicles' convergence became an injury.
- After the impact the light vehicle was pushed 18 m down the ramp and rolled onto its left side, which is how the collision produced the driver's injuries.

Other factors that increased risk

- Only one emergency response paramedic was rostered on night shift, with the second responder called in from the village, and it is very likely that this arrangement increased risk at the mine, although the post-incident review found that the resulting delay did not affect this occurrence or its outcome. (Safety issue)
- The truck operator was on night 10 of 14, in the eleventh hour of a 12.5 hour night shift and in the circadian low (the period of lowest natural alertness), and it is likely that this roster pattern increased risk for operators on the pit, although the operator's recorded response showed no sign that it played a part in this occurrence.

Other findings

- The investigation examined whether the ramp surface was wet and the truck could not stop in the distance available, and ruled this out because the ramp was dry and the truck achieved its tested braking performance.
- Whether the haul truck's service brakes were degraded and could not achieve the required stopping performance was also examined and ruled out, because the brakes met specification in tests before and after the occurrence and during the occurrence itself.
- The truck operator braked and steered into the windrow within 1.4 seconds of sighting the light vehicle, which reduced the impact speed from 37 km/h to 24 km/h and kept the collision against the inner side of the ramp rather than the open edge.

What is being done

Traffic management guidance (MSG-14). The guidance did not define what counts as a significant change to the road network. It also left each site to nominate its own interaction zones and to choose its own means of making a light vehicle detectable. No action in response to this issue has been recorded. [NOT ESTABLISHED: any action taken or proposed by the State Mines Inspectorate or the operator in response to this issue]

Spur opened under a works order with no management of change. The works order exclusion removed the management of change record, and with it the referrals to the plan owner, the survey department and the contractor's technical services group. No action in response to this issue has been recorded. [NOT ESTABLISHED: any action taken or proposed, and its status]

Contractor induction accepted as equivalent. The access card system holds no field for the site supplement or the pit driving assessment, so the verification the standard requires had nowhere to happen. No action in response to this issue has been recorded. [NOT ESTABLISHED: any action taken or proposed, and its status]

Positive radio calls only at listed intersections. On the night of the occurrence the site issued a direction requiring a positive call at every Stage 4 intersection, listed or not. The investigation confirmed that the list could not be updated between annual reviews, and raised this with the Mining Manager ahead of the draft report. The investigation intends to make a recommendation on this issue in its own right. [NOT ESTABLISHED: whether the overnight direction remains in force, and any permanent change to the plan]

Collision awareness alerts suppressed inside an extended geofence. No action in response to this issue has been recorded. [NOT ESTABLISHED: any action taken or proposed, and its status]

Single paramedic on night shift. The investigation has asked the emergency response coordinator about the roster on the site's other three night shifts. This matter remains open until that answer is received. No action in response to the roster arrangement has been recorded.

Distribution

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